

Underwater Vision-Based Surveillance Using ML and AI

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Day 8 — Dataset & Annotation Quality Report

1. Objective

Audit the three supplied underwater datasets (Well, DIATAquarium, Aquatic Plant) for:

- Image/video files, labels/targets, duplicates and split integrity
- Resolution, class/scene coverage, visibility and corrupt samples
- Annotation/pair consistency
- Preprocessing and annotation corrections needed before training

2. Dataset Summary

Dataset	Classes	Train	Valid	Test
DIATAquarium	11	8,121	773	Not present locally
Aquatic Plant	1	1,892	179	89
Well	4	3,795	191	184

3. Image and Label Audit

- All three datasets contain image data; no separate video files were identified.
- YOLO annotation format was checked across all datasets.
- 0 invalid YOLO annotation entries were found (all label rows had 5 valid fields).
- Aquatic Plant has 13 training images without matching label files.
- Well has 15 training images without matching label files.
- Empty labels were found and may represent legitimate background/negative images, so they were not automatically deleted.

4. Resolution and Corrupt-Sample Check

A sample of 50 images per available split was checked for resolution consistency and file corruption.

Dataset / Split	Resolution
DIATAquarium — train	1920 × 1080
DIATAquarium — valid	2048 × 1536
Aquatic Plant — all splits	640 × 640

Dataset / Split	Resolution
Well — all splits	640 × 640

- No corrupted images were found in the checked samples.
- DIATAquarium's train/valid resolution mismatch can be handled during preprocessing (resizing/normalisation).

5. Class and Scene Coverage

- DIATAquarium: all 11 classes are present, but class frequencies are uneven.
- Aquatic Plant: contains a single class — Aquatic_plant.
- Well: severe class imbalance — fishes: 50,194, school-of-fish: 653, stone: 5, Inlet-pipe: 3.
- Scene/lighting distribution was not fully quantified and remains a pending check.

6. Visibility and Image Quality

Sampled images showed the following degradation patterns:

- Haze
- Low contrast
- Reduced visibility
- Color variation / distortion.

7. Annotation Consistency

Three sampled training images from each dataset were visually reviewed by drawing bounding boxes from their label files.

- Bounding boxes generally matched visible objects.
- Class labels appeared consistent with the objects shown.
- No obvious major annotation errors were observed in the sample.
- A larger annotation review is recommended before final training, since only 3 images per dataset were checked.

8. Duplicate and Split Integrity

- Exact duplicate checking was performed on samples of up to 100 images per split.
- No exact cross-split duplicates were found in the checked samples.
- 6 within-split duplicates were found in the Aquatic Plant training sample.
- Full duplicate/near-duplicate checking is recommended before final split freezing.

Temporal/source (video-frame) leakage was not fully checked during the Day 8 audit and remained a pending risk for later preprocessing and split preparation.

9. Important Dataset Issue — DIATAquarium

- No local test folder is present for DIATAquarium.
- The local data.yaml points both val and test paths to test/images.
- Split handling will be finalized before training, according to mentor guidance.

10. Correction Plan

Issue	Proposed Correction
Different resolutions	Standardize image size during preprocessing
Haze / low contrast	Evaluate White Balance + CLAHE
Missing labels	Review images individually before correction/removal
Empty labels	Retain unless confirmed incorrect
Duplicate images	Review confirmed duplicates before training
Class imbalance	Consider suitable sampling/balancing strategies
Annotation errors	Correct only confirmed errors
DIAT YAML issue	Resolve paths according to final split strategy
Pending quality checks	Extend important checks before final training